Prince Abu, FRM

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Summary:

- Quantitative Economist and FRM-Certified with expertise in financial risk management, econometric modeling, predictive analytics, machine learning, and statistical programming.
- Experienced in developing and validating quantitative models, analyzing large-scale financial and macroeconomic datasets, performing forecasting, scenario analysis, and model validation using Python, SQL, MATLAB, R, STATA, and SAS.
- Strong communicator with experience collaborating across teams and translating complex data into insights for both technical and non-technical audiences.

Education:

Ph.D. Quantitative Economics, **Minor in Econometrics** – Middle Tennessee State University (Expected May 2027)

- Research Focus: macroeconomics, econometrics, financial risk analysis.

M.A. Economics – Middle Tennessee State University GPA: 3.95
Master of Business Administration – Regenesys, South Africa Distinction
B.S.c Economics and Statistics – University of Benin GPA: 3.74

Relevant Experience:

Research Assistant – Middle Tennessee State University (Aug. 2022 – Present)

- Process and analyze 20+ years of macroeconomic data using Python, MATLAB, SAS, and Excel to estimate structural relationships and forecast economic outcomes. Utilized data visualization tools for presentations.
- Conduct extensive sensitivity and scenario analyses across alternative policy assumptions to evaluate model robustness under economic uncertainty.
- Apply Bayesian and maximum likelihood estimation techniques to calibrate and validate economic models.
- Executed 1000+ Monte Carlo simulations to evaluate model sensitivity under alternative economic scenarios.
- Develop optimization and numerical algorithms to estimate and solve structural models, executing 1,000+ scenario simulations to evaluate macroeconomic uncertainty and model robustness.
- Collaborate with faculty researchers and interdisciplinary teams to develop quantitative models, validate empirical results, and translate research findings into actionable economic insights.

Economist/Financial Analyst – Tonmax Engines (Dec. 2020 – April 2022)

- Developed and launched the company's website, increasing customer inquiries by 15% and contributing to a 5% growth in sales through an improved digital customer experience.
- Analyzed large-scale transactional and financial datasets to identify cost drivers, contributing to a 15% reduction in operating expenses.
- Built regression and time-series models that improved demand forecast accuracy by 12%, enhancing inventory optimization and operational efficiency.
- Developed statistical forecasting and risk models to project revenue, quantify business uncertainty, and support financial planning.
- Performed scenario and sensitivity analyses to evaluate the financial impact of changing market conditions and operational risks.
- Automated financial reporting workflows and executive dashboards, reducing reporting time by 80% while improving data quality and decision support.

Technical Skills:

- Programming: MATLAB, Python, R, Stata, SAS, VBA.
- Data Visualization: Power BI, Tableau, SQL, ArcGIS Pro
- Modeling: Market risk (VaR, CVaR, Monte Carlo Simulation, GARCH); credit risk (Credit Scoring, Credit Portfolio), operational risk, enterprise risk (stress testing, scenario analysis), machine learning.

Professional Certifications:

- CFA Level 1 Candidate
- Financial Risk Manager (FRM) Certified by Global Association of Risk Professionals (09/2026)
- Google GCP Professional Machine Learning Engineer (01/2026)
- Google GCP Professional Data Engineer (11/2025)
